

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No.
18 UME 21116

B.E. II (M)

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Maths (M)

SECOND B.E.

MECHANICAL ENGINEERING

III SEMESTER EXAMINATION - 2019

MA - 205 A: MATHEMATICS

Time - Three Hours

Maximum Marks - 100

Note:- (1) Answer any **Five** questions.

(2) All questions carry equal marks.

SECTION - A

1. (a) Solve the equations :

$$\frac{dx}{\cos(x+y)} = \frac{dy}{\sin(x+y)} = \frac{dz}{z}$$

(b) Solve:

$$(y^2 + yz)dx + (xz + z^2)dy + (y^2 - xy)dz = 0$$

2. (a) Solve

$$(y^2 + z^2 - x^2)p - 2xyq + 2xzr = 0$$

(b) Solve the equation by Charpit's method :

$$(p^2 + q^2)y = qz$$

3. (a) Solve:

$$r - 2s + t = \sin(2x+3y)$$

(b) Solve by Monge's method :

$$x^2 r + 2xy s + y^2 t = 0$$

SECTION - B

4. (a) By using Newton - Gregory forward difference interpolation formula, compute $y(3.62)$ from the following table

x	:	3.60	3.65	3.70	3.75
y	:	36.598	38.475	40.447	42.521

(b) Apply Lagrange's formula to find $f(5)$, given that

$$f(1)=2, f(2)=4, f(3)=8, f(4)=16, f(7)=128$$

5. (a) Integrate the function $f(x) = \frac{1}{1+x}$ over $[0,1]$ by Simpson's rule. Thus find the value of $\log_e 2$.

(b) By using Newton - Raphson method, find the root of $x^4 - x - 10 = 0$ which is near to $x = 2$ correct to three places of decimal.

(a) Find the solution of the following system of equations by Gauss-Jordan reduction method :

$$10x_1 + x_2 + x_3 = 12$$

$$x_1 + 10x_2 + x_3 = 12$$

$$x_1 + x_2 + 10x_3 = 12$$

(b) Solve the equation

$$\frac{dy}{dx} = x + y$$

with initial condition $y(0) = 1$ by Runge - Kutta method from $x = 0.1$ to $x = 0.4$ with $h = 0.1$

SECTION - C

7. (a) Find the Laplace transform of $\frac{\sin at}{t}$. Does the transform of $\frac{\cos at}{t}$ exist?

(b) Find the inverse Laplace transform of $\frac{1}{s^2(s^2 - a^2)}$

by using the convolution theorem.

8. (a) Solve :

$$(D^2 + D)y = t^2 + 2t \quad \left(D = \frac{d}{dt} \right)$$

where $y(0) = 4, y'(0) = -2$.

(b) Find the Fourier transform of

$$f(x) = \begin{cases} x, & |x| \leq a \\ 0, & |x| > a. \end{cases}$$

8. (a) Apply Laplace transform to solve

$$\frac{d^2 y}{dt^2} + y = 6 \cos 2t, \text{ given that}$$

$$y = 3, \frac{dy}{dt} = 1 \text{ when } t=0$$

(b) find the Fourier Cosine transform of e^{-x^2}

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Roll No.

B.E. II (M)

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Maths (M)

SECOND B.E.

MECHANICAL ENGINEERING
III SEMESTER EXAMINATION-2018
MA-205A: ADVANCE ENGINEERING
MATHEMATICS

Time - Three Hours
Maximum Marks - 100

Note :- (1) Attempt any Five questions.

(2) All question carry equal marks.

Section-A

1. (a) Solve the Simultaneous equation.

$$\frac{dx}{dt} + 5x + y = e^t$$

$$\frac{dy}{dt} - x + 3y = e^{2t}$$

(b) $yz^2(x^2-yz)dx + x^2z(y^2-xz)dy + xy^2(z^2-xy)dz=0$

~~2.~~ (a)

Solve:

$(y^3x - 2x^4)p + (2y^4 - x^3y)q = 9z(x^3 - y^3)$

(b)

Solve by Charpit's method:

$2xz - px^2 - 2qxy + pq = 0$

3. (a)

Solve:

$4 \frac{\partial^2 z}{\partial x^2} - 4 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 16 \log(x+2y)$

(b)

Solve by Monge's method:

$t - r \sec^4 y = 2q \tan y$

Section-B

4. (a)

Show that

$\frac{\Delta^2}{E} \sin(x+h) + \frac{\Delta^2 \sin(x+h)}{E \sin(x+h)} = 2(\cosh-1) \{ \sin(x+h) + 1 \}$

h being the differencing interval.

(b) By Using Lagrange's interpolation formula, find

the form of the functions given by

x	0	1	2	5
f(x)	2	3	12	147

5. (a)

Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using simpson's $\frac{1}{3}$ rule. Hence

obtain the approximate value of π .

(a)

The equation $x^6 - x^4 - x^3 - 3 = 0$ has one real root between 1.5 and 1.6. Find the exact root to four decimals by False Position method.

~~6.~~ (a)

Use Runge-Kutta method to solve.

$\frac{dy}{dx} = xy$

for $x=1.4$ Initially $x=1, y=2$ (taking $h=0.2$)

(b)

Solve the system of equations

$27x + 6y - z = 85$

$6x + 15y + 2z = 72$

$x + y + 54z = 110$

By Gauss-seidal method.

Section-C

7. (a)

Find the Laplace transform of $F(t)$ where

$$F(t) = \begin{cases} t & 0 \leq t \leq \frac{1}{2} \\ t-1 & \frac{1}{2} \leq t \leq 1 \\ 0 & t > 1 \end{cases}$$

(b) Use the convolution theorem to find

$L^{-1} \left\{ \frac{p}{(p^2+4)^2} \right\}$

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Roll No. 14M... 385 (5)

BE.II(M)

Maths (M)

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SECOND B.E. MECHANICAL ENGINEERING

III SEMESTER EXAMINATION - 2017

MA-205 A: ADVANCE ENGINEERING

MATHEMATICS

Time Allowed - Three Hours

Maximum Marks : 100

Note:- (1) Attempt any FIVE questions.

(2) all questions carry equal marks.

1. (a) Solve the Simultaneous equation-

$$t \, dx = (t-2x) \, dt$$

$$t \, dy = (tx+ty+2x-t) \, dt$$

(b) Solve-

$$(y^3x-2x^2)p + (2y^4-x^3y) \, q = 9 \, z \, (x^3-y^3)$$

2. (a) Solve.

$$(2xz-yz) \, dx + (2yz-xz) \, dy - (x^2-xy+y^2) \, dz = 0$$

(b) Solve.

$$z = pqxy \text{ by charpit's method.}$$

3. (a) Solve.

$$q^2r-2pqrs+p^2t=0 \text{ by Monge's Method.}$$

8231

(Contd.)

(b) Solve by the method of Separation of variables.

$$\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}, u(0, y) = 8e^{-3y}$$

10

4. (a) The ordinates of the normal curve are given by the following task.

x: 0.0 0.2 0.4 0.6 0.8

y: 0.3989 0.3910 0.3683 0.3332 0.2897

find (i) y(0.25) (ii) y(0.62)

10

(b) (i) Evaluate

$$\Delta^2 \left[\frac{5x+12}{x^2+5x+6} \right]$$

(ii) Prove that

$$\Delta \log f(x) = \log \left[1 + \frac{\Delta f(x)}{f(x)} \right]$$

5

5. (a) Use Bessel formula to find y_{25} given that

$y_{20} = 24, y_{22} = 32, y_{28} = 35, y_{30} = 40$

10

(b) Find the value of $\log_2 2$ from $\int_0^1 \frac{x^2}{1+x^3} dx$

10

using Simpson's 1/3 rule, taking four equal intervals.

6. (a) Using Runge-kutta method of fourth order, find an approximate value of $y(0.2)$ by solving

$$\frac{dy}{dx} = -2xy^2 \text{ with } y(0) = 1$$

(b) Solve the following system of equations using Gauss Jordan elimination method

$$2x + y + z = 10, 3x + 2y + z = 8, x + y + z = 6$$

8231

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(Contd.)

$$x+4y+9z=10$$

$$L \left\{ \frac{\sin^2 t}{t} \right\} = \frac{1}{4} \log \left(\frac{s^2+4}{s^2} \right)$$

Hence deduce that

$$\int_0^\infty \frac{\sin^2 t}{t^2} dt = \frac{\pi}{2}$$

(b) Find

$$L^{-1} \left[\frac{s^2}{s^4+4a^4} \right]$$

8. (a) Solve

$$(\Delta^2+9)y = \cos 2t$$

given that $y(0)=1, y(\pi/2)=-1$

(b) Find the Laplace transform of

$$f(x) = \begin{cases} x^2 & ; |x| \leq a \\ 0 & ; |x| > a \end{cases}$$

8231

Q.8. (a) Prove that

$$\nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)$$

$$\text{where } Y^2 = x^2 + y^2 + z^2$$

(b) Verify stoke's theorem for the function

$$F = z\mathbf{i} + x\mathbf{j} + y\mathbf{k} \text{ where } C \text{ is the unit circle in the}$$

xy-plane bounding the hemisphere

$$Z^2 = 1 - x^2 - y^2$$

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PSLP 5511/24

Roll No.

B.E. II (M)
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Maths (M)

SECOND B.E. EXAMINATION (2016)
(MECHANICAL ENGINEERING)
(III SEMESTER EXAMINATION SCHEME)
MA-205 A: MATHEMATICS (M)

Time Allowed - Three Hours
Maximum Marks - 60

Note : (1) Attempt any FIVE questions.

(2) All questions carry equal marks.

(Contd.)

Q1.(a) Evaluate

$$\iiint xyz \, dx \, dy \, dz$$

Where the region of integration is the volume of the ellipsoid in the positive octant

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1$$

(b) Evaluate

$$\int_0^a \int_0^x \int_0^{x+y} e^{x+y+z} \, dx \, dy \, dz$$

Q.2. (a) Solve the simultaneous equation-

$$\frac{dx}{dt} + 2x = t, \quad \frac{dy}{dt} - 2x = \frac{1}{t}$$

(b) Solve-

$$\frac{dx}{\cos(x+y)} = \frac{dy}{\sin(x+y)} = \frac{dz}{z}$$

Q.3. (a) Solve

$$z^2(p^2 + q^2) = x^2 + y^2$$

(b) Solve $Z = pq$ by Charpit's method

KK/pg2/400

Q.4. (a) Solve $t + S + q = 0$

(b) Solve $yt - q = yx$

Q.5. (a) Prove that

$$(i) P_n(1) = \frac{1}{2} n(n+1)$$

$$(ii) P_n(-1) = (-1)^{n-1} \frac{n(n+1)}{2}$$

(b) Prove that

$$nP_n = x P_n' - P_{n-1}'$$

Q.6. Prove that

$$(a) x J_n'(x) = n J_n(x) - x J_{n+1}(x)$$

(b) Define and prove the orthogonal property of Bessel function of first kind.

Q.7. (a) Evaluate $\int_0^1 \left[\log \frac{p^2 + b^2}{p^2 + a^2} \right]$

(b) Solve $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + y = x$

given that $y(0) = -3$, $y(1) = -1$ by Laplace transform

KK/pg3/400

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Roll No. ... 3200056...

B.E. II (M)

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Maths (M)

Second B.E. Examination, 2014
(Mechanical Engineering)
(III - Semester Examination Scheme)

MA-205 A : MATHEMATICS (M)

Time : Three Hours
Maximum Marks : 60

- Note : (1) Attempt any **FIVE** questions
(2) All questions carry equal marks.

1.(a) Solve:

$$\frac{dx}{x^2 - yz} = \frac{dy}{y^2 - zx} = \frac{dz}{z^2 - xy}$$

(b) $(2xz - yz) dx + (2yz - xz) dy - (x^2 - xy + y^2) dz = 0$

2.(a) Solve:

$$\left(\frac{y - z}{yz} \right) p + \left(\frac{z - x}{zx} \right) q = \left(\frac{x - y}{xy} \right)$$

(b) Solve the equation by Charpits method

$$(p^2 + q^2) y = qz$$

3.(a) Solve: $r + (a+b)s + abt = xy$

(b) Solve: $r - 2s + t = \sin(2x+3y)$

4.(a) Prove that:

$$P_n(x) = \frac{1}{2^n n!} \left[\frac{d^n}{dx^n} (x^2 - 1)^n \right]$$

(b) Prove that: $(2n+1)xP_n = (n+1)P_{n+1} + nP_{n-1}$

5.(a) Prove that

$$\int_{-1}^1 (x^2 - 1)P_{n+1}'P_n' dx = \frac{2n(n+1)}{(2n+1)(2n+3)}$$

(b) Prove that $J_n(x)$ is equal to the coefficient of Z^n in the expansion of $\exp \left[\frac{x}{2} \left(z - \frac{1}{z} \right) \right]$ when in

is a positive integer, in ascending power of Z .

6.(a) Prove that: $xJ_n'(x) = nJ_n(x) - xJ_{n+1}(x)$

(b) Prove that: $\sqrt{\frac{\pi x}{2}} J_{3/2}(x) = \frac{\sin x}{x} - \cos x$

7.(a) Using the Laplace transforms, find the solution of the initial value problem

$$y'' + 2y' + 5y = e^{-t} \sin t$$

$$y(0) = 0, y'(0) = 1$$

(b) Find the Laplace inverse of $\frac{s^3 + 6s^2 + 14s}{(s+2)^4}$

8.(a) If $\vec{A} = 2x^2\vec{i} - 3yz\vec{j} + xz^2\vec{k}$ and $f = 2z - x^3y$

Find (i) $\vec{A} \cdot \nabla f$ and (ii) $\vec{A} \times \nabla f$ at the point $(1, -1, 1)$

(b) Verify Stokes theorem for

$$\vec{A} = xz\vec{i} - yz\vec{j} + x^2yz\vec{k}$$

where S is the surface of the region bounded by $x=0, y=0, z=0, 2x+y+2z=8$ which is not included in the xz plane.

$$2x + y + 2z = 8$$

$$x^2yz = x^2yz$$